

Nouman Rasheed

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SUMMARY

Senior AI Engineer with 4+ years building production Generative AI systems: multi-agent LLM platforms (LangGraph, Claude, GPT, Gemini), enterprise RAG, and custom generative models for handwriting synthesis trained from scratch. Founder-engineer at an American AI startup, enterprise GenAI implementations in Saudi Arabia, with a strong Computer Vision and Edge AI background.

PROFESSIONAL EXPERIENCE

Ink AI

Delaware, USA (Remote)

Founding AI Engineer

June 2024 - Present

- Founding Engineer at a ink.ai startup Founded by Rich Miner (co-founder of Android). Designed the first stack from scratch, including the Android app, Kotlin Ktor/gRPC back-end microservices, GCP infrastructure, and ML research.
- Designed and trained generative handwriting synthesis models from scratch: Cursive Transformers with cross-attention, encoder-decoder models with Gaussian memory masking and MDN outputs, GPT-2-style decoders with polar-coordinate stroke tokenization, and VRNNs on CASIA, IAM, and custom datasets grown to 1M+ samples, running 200+ tracked experiments in W&B.
- Shipped the V2 cursive transformer (trained on 650K samples) as a production FastAPI synthesis service with KV-cache fp16 inference, coalesced GPU batching, variant rotation, and a Postgres result cache.
- Shipped production AI features, fine-tuned GPT models for entity extraction, text prediction, placeholder completion, and tag generation RAG-powered meeting briefs with attendee enrichment via LinkedIn, Apollo, and SerpAPI executive summaries synthesized from handwritten notes and voice transcripts built a prompt-management layer with frontmatter-based model routing (OpenAI/Gemini) and an automated prompt-eval test harness.
- Implemented Gemini tool-calling agents orchestrating Gmail, Calendar, and Docs APIs for automated notebook generation built real-time voice transcription, multi-mode text refinement, and dual spell checking.
- Built a data collection web app with validation pipeline and curated a proprietary handwriting dataset of 50,000+ samples from 450+ writers engineered training-data pipelines with Gemini-based stroke alignment and multi-engine character segmentation (Gemini, Google HWR, Apple Vision) plus synthetic word composition.
- Architected multi-region GCP infrastructure using Cloud Run, Terraform IaC, Cloud Build CI/CD, and Cloud SQL built a RAG pipeline with Vertex AI embeddings and PGVector, image retrieval with FAISS and ViT-L/14, and DynamoDB caching with TTL-based invalidation.
- Led development of 3 Kotlin Ktor gRPC microservices (auth, AI, notes) and a Kotlin Android app with Jetpack Compose and offline-first Room DB defined Protobuf services and contracts across the stack.

PrimeGate

Riyadh, Saudi Arabia

AI Lead Engineer

November 2024 - Present

- Architecting the company's Agentic Platform v2: a LangGraph supervisor graph that decomposes queries into sub-tasks dispatched in parallel to runtime user-defined agents, with synthesizer and critic stages, per-tier Claude model routing, and a Docling-based document ingestion service over Postgres, Qdrant, and MinIO.
- Multi-Agent RAG Platform (internal pilot with executive users): engineered an on-premises multi-agent system using LangGraph ReAct orchestration with 11 specialized agents with bilingual Arabic/English semantic search over policies, Dynamics 365 entities, and knowledge bases backed by Qdrant - deployed fully on-premises as 9 containerized microservices with Azure AD OAuth.
- Building PrimeTender, a multi-tenant Saudi tender-intelligence SaaS: Claude-powered go/no-go bid analysis with budget and readiness scoring, a 4-agent quick-bid PDF orchestrator, pgvector with Voyage embeddings for bilingual semantic search, and per-call LLM cost tracking.
- Designed Text-to-SQL generation with role-based access control (Employee, Manager, HR tiers) over Dynamics 365 HR data for secure querying of compensation, attendance, and leave data.
- Building SOTA Access Control (face recognition live in production): ArcFace-based employee identification across concurrent video streams, with ANPR and multi-object tracking in active development.

Ciena (Contractor)

Remote, Canada

Computer Vision Engineer

September 2024 - August 2025

- Built an end-to-end computer vision system to analyze fiber network chassis images - detecting chassis, cards, ports, and fiber anomalies for Ciena network equipment deployed in large-scale META data center environments.

- Delivered the service as a FastAPI application with Redis-backed async batching on AWS SageMaker with auto-scaling.
- Integrated YOLOv8 for chassis, card, and port segmentation, fiber-kink detection models, and a UNet ONNX model for fiber mask extraction applied INT8 quantization via ONNX Runtime for inference efficiency.
- Designed a JSON-driven port mapping algorithm to handle occlusion and synthesize missing detections and implemented bend-radius estimation validated against ITU G.652/G.657 standards.
- Integrated Llama 3.2 Vision for shelf and fiber label OCR built best-of-3 frame selection workflow for improved detection reliability.

DEER

Senior AI Engineer

Dammam, Saudi Arabia

May 2022 - November 2024

- Deployed real-time forklift and person detection with deployment on Google Coral TPU (MobileNet-v2) and Nvidia Jetson Orin (YOLOv8 with custom DeepSORT appearance-based tracking) for industrial safety monitoring at production-grade accuracy.
- Built a facial recognition pipeline using LFFD, FaceNet, and DeepStream, and a Saudi license plate recognition system using LPRNet, deployed as part of site security infrastructure.
- Developed a healthcare chatbot proof-of-concept using fine-tuned LLaMA 3.1 with GraphRAG document retrieval over Neo4j built an LLaMA 3.2-based industrial plant report automation POC (Aramco pilot engagement).
- Contributed to Arabic ASR experimentation, including tokenizer evaluation and preprocessing work for Arabic speech models.

Sigma Vendors

Computer Vision Engineer

Lahore, Pakistan

September 2021 - April 2022

- Implemented a Raspberry Pi 3 B+ hospital door entry system using Tiny YOLOv5 for real-time edge inference during COVID-19 screening.
- Built a mask detection and people counting system with an interactive Streamlit monitoring dashboard.

EDUCATION

University of Kotli (AJK)

Bachelor of Science in Computer Science

Kotli, Pakistan

October 2017 - November 2021

- Final-year project: Intelligent waste management system using custom Mask R-CNN for multi-class waste classification. Selected as first team from the university to present at MUST Industrial Exhibition.

TECHNICAL SKILLS

AI/ML Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, ONNX Runtime, DeepStream, Weights & Biases

LLMs & Agents: Claude (Anthropic), GPT, Gemini, LLaMA (QLoRA fine-tuning), Qwen, LangGraph supervisor/ReAct agents, LangChain, LlamaIndex, RAG, GraphRAG, Text-to-SQL, tool calling, prompt engineering & automated prompt evals

Generative Models: Transformer sequence generation (MDN heads, KV-cache inference), VRNNs, diffusion models, custom tokenization schemes

Computer Vision: YOLOv8, ArcFace, FaceNet, OpenCV, UNet, Mask R-CNN edge deployment on Jetson Orin, Coral TPU, Raspberry Pi

NLP & Search: Arabic/English bilingual embeddings, Sentence Transformers, Vertex AI & Voyage Embeddings, Qdrant, PGVector, FAISS, Meilisearch, ChromaDB

Infrastructure & Cloud: Docker, Kubernetes, Terraform, AWS (SageMaker, S3, Lambda), GCP (Cloud Run, Cloud Build, Cloud SQL), Nginx, Redis, DynamoDB

Backend & APIs: Python, Kotlin, FastAPI, Flask, gRPC/Protobuf, OAuth 2.0 (Azure AD), PostgreSQL, Neo4j

SELECTED PROJECTS

MidadaAI - Arabic E-Commerce Content Engine | *Side Project*

- AI-powered SaaS platform for Arabic e-commerce content generation using GPT-4, with Salla/Zid store connectors, Moyasar payments integration, and Inngest background processing. Stack: Next.js 14, tRPC, Drizzle ORM, Supabase, Clerk.

Handwriting Generation with Transformers | *Research*

- Trained Style-Disentangled Transformer models on handwriting datasets to generate realistic handwriting from text input, with configurable style priming and checkpoint-based generation. Stack: Python, PyTorch.

Business Intelligence Dashboard | *Personal Project*

- Campaign management BI suite with ETL pipelines, ARIMA time-series forecasting, Gemini Pro prescriptive analytics, and multi-view dashboards (Gantt, budget, KPI). Stack: Django, ReactJS, Docker.